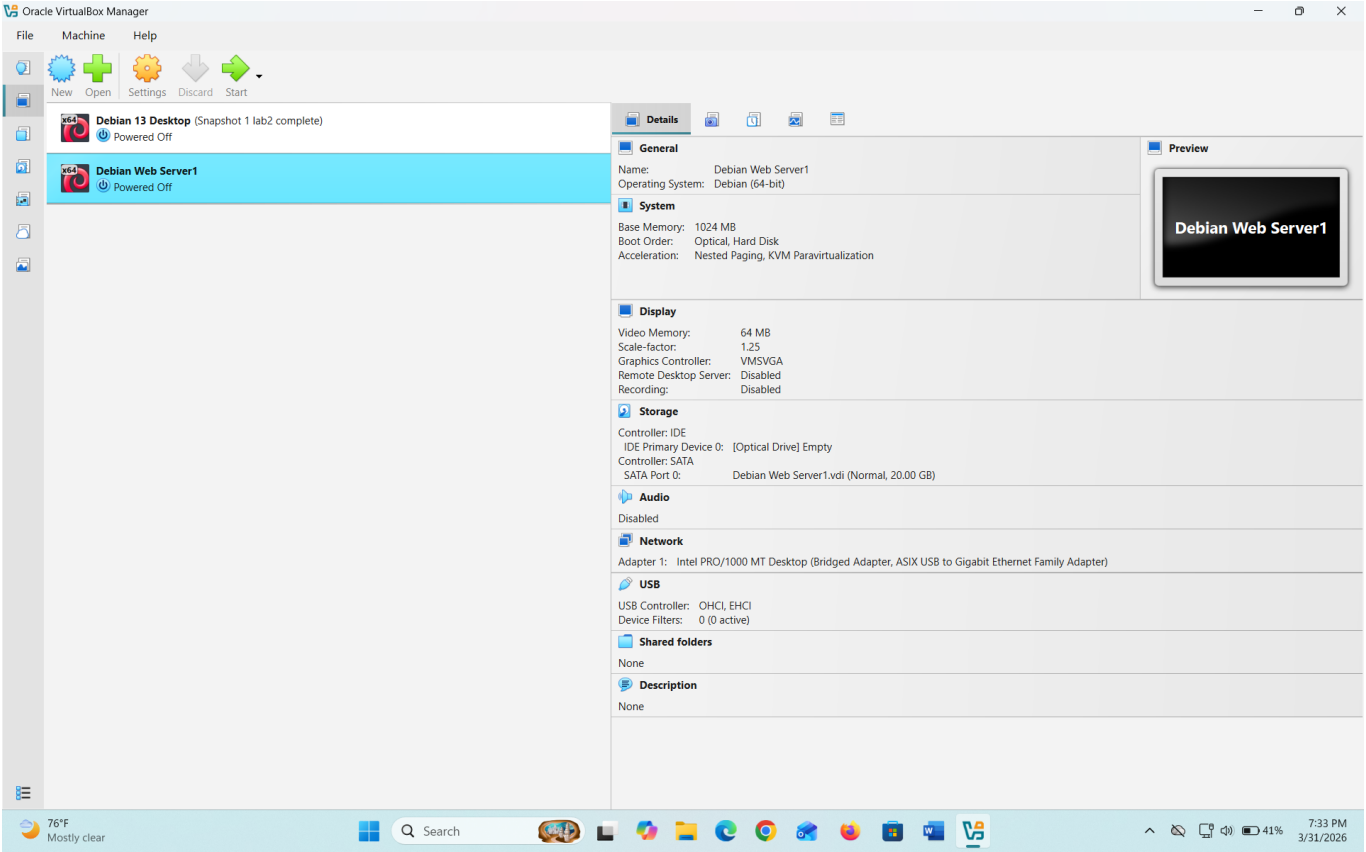
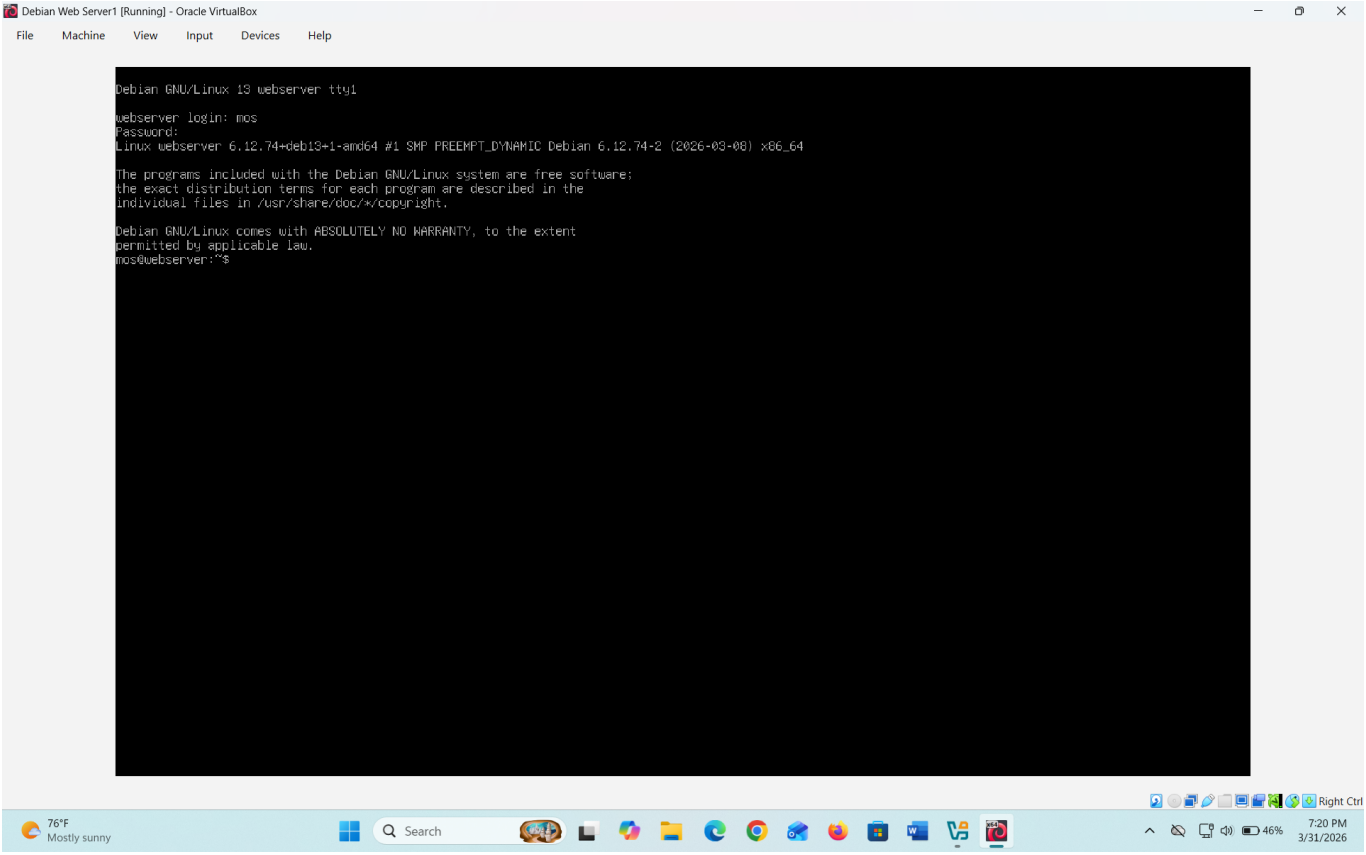
Deliverable 2 Setting up the Web Server

What are the server hardware specifications (virtual machine settings)? Take a screenshot - don’t type it!**



What is the Debian Login Screen? (Take screenshot - do not type it!)



What is the IP address of your Debian Server Virtual Machine? (type the command and show a screenshot of the commands output)

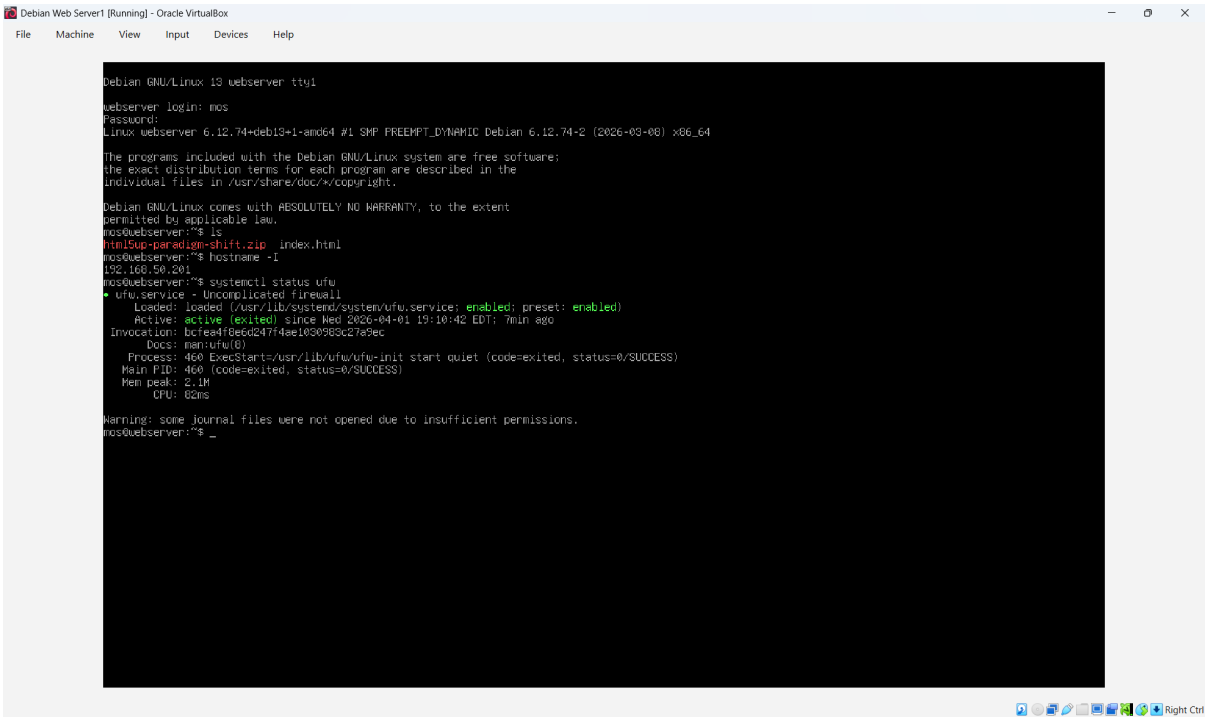
`hostname -I`

```
Debian GNU/Linux 13 webserver tty1

webserver login: mos
Password:
Linux webserver 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
mos@webserver:~$ hostname -i
127.0.1.1
mos@webserver:~$ hostname -I
192.168.50.201
mos@webserver:~$
```

How do you work with the Firewall in Debian? (Type and explain what each command does)

- **Check UFW (Uncomplicated Firewall)** UFW is a common firewall tool on Debian (especially for beginners).

- **Formula** `sudo ufw status`



- **What it does:**
  - Shows whether UFW is **active** or **inactive**
  - Lists current firewall rules (allowed/denied ports)

- **Check systemd service for UFW**

- **Formula**
  - `sudo systemctl status ufw`
  - **What it does:**
    - Checks if the UFW service is running in the background
    - Shows details like:
    - Active state (running/stopped)
    - Logs
    - When it started

- **Check iptables (low-level firewall)**
  - Even if UFW isn't used, Debian may use iptables.
  - **Formula**
    - `sudo iptables -L`
  - **What it does:**
    - Lists all firewall rules in the kernel
    - Shows chains like:
    - INPUT (incoming traffic)
    - OUTPUT (outgoing traffic)
    - FORWARD (routing traffic)
    - If rules are listed, firewall filtering is happening.

- **Check nftables (modern replacement for iptables)**

    - Newer Debian versions use nftables.
    - **Formula**
    - `sudo nft list ruleset`
    - **What it does:**
        - Displays all nftables firewall rules
        - Shows tables, chains, and filtering rules

- **Check if nftables service is running**

    - **Formula**

    - `sudo systemctl status nftables`

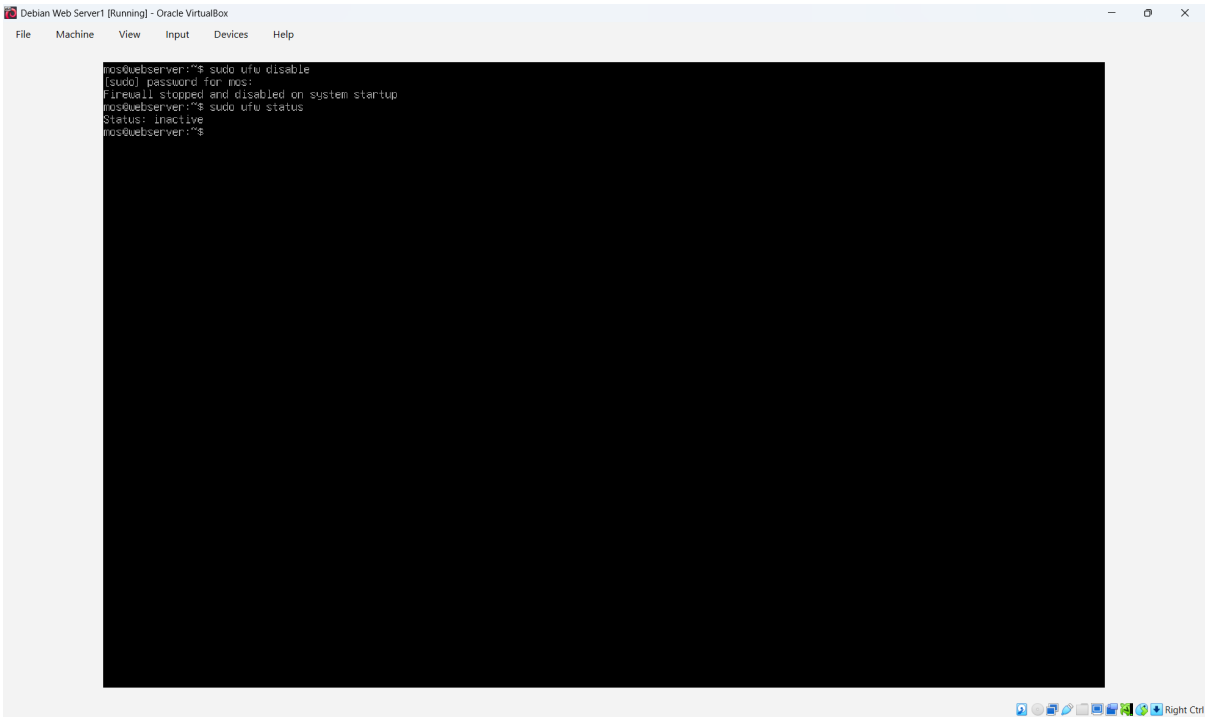
    - **What it does:**

        - Confirms whether nftables is active
        - Shows service status and logs

- **How do you check if the Firewall is running?**

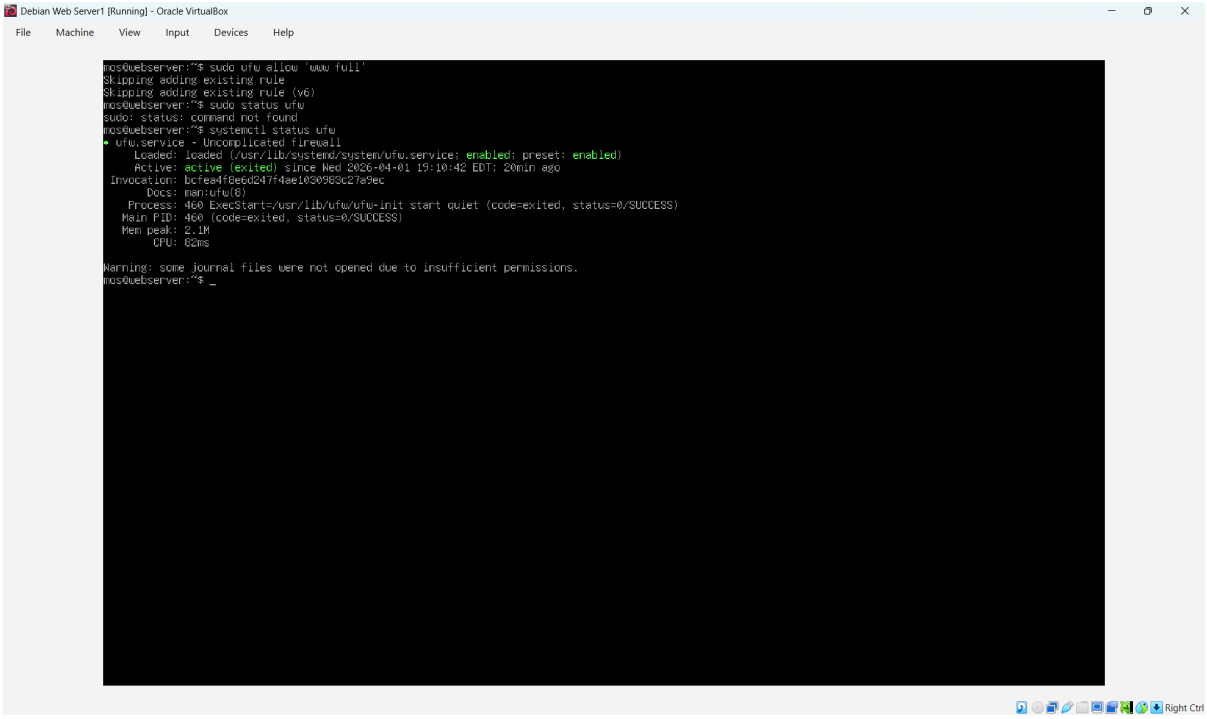
    - By using the command `systemctl status ufw`

- **How do you disable the Firewall?**

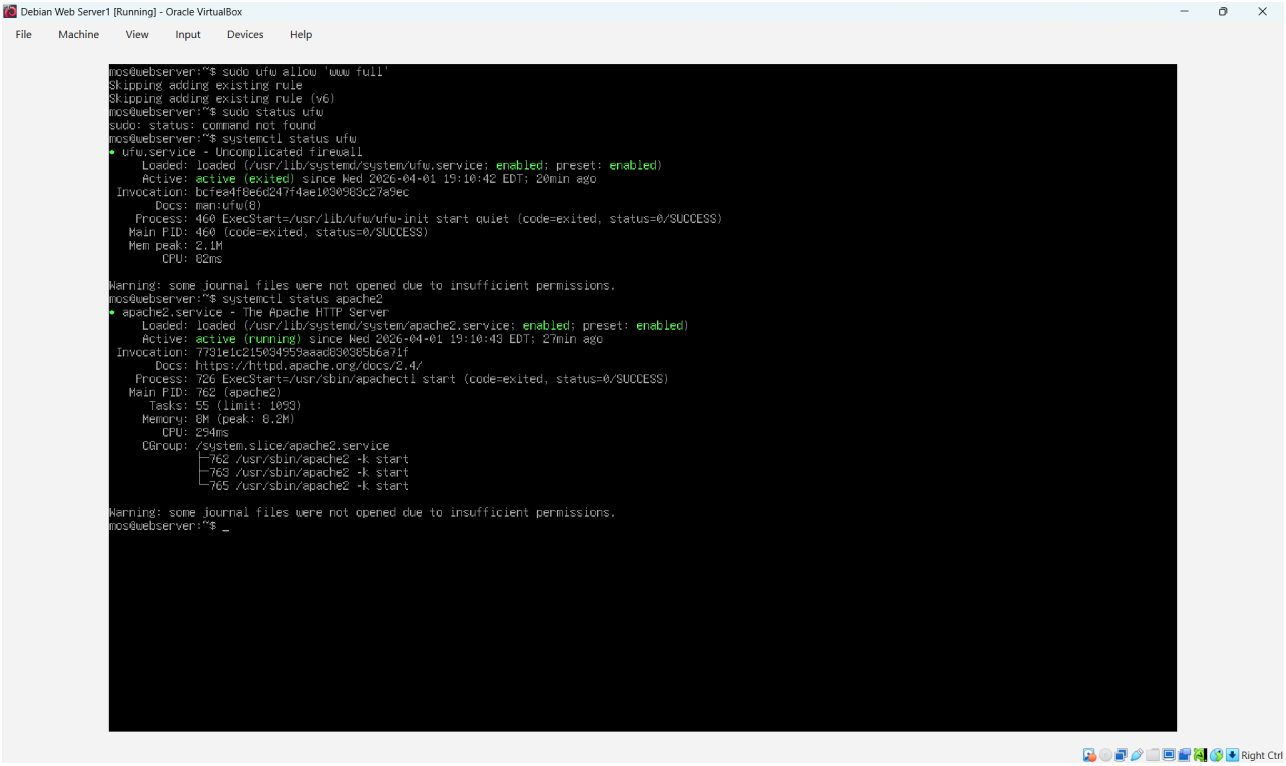
    - By using the command `sudo ufw disable`
    - 

- **How do you add Apache to the Firewall?**

    - By using the command `sudo ufw allow 'Apache Full'`

- 

What different commands do we use to work with Apache? (Type and explain what each command does and include a screenshot!)

- **What is the command you use to check if Apache is running?**
- By using the command `system status apache2`
- 

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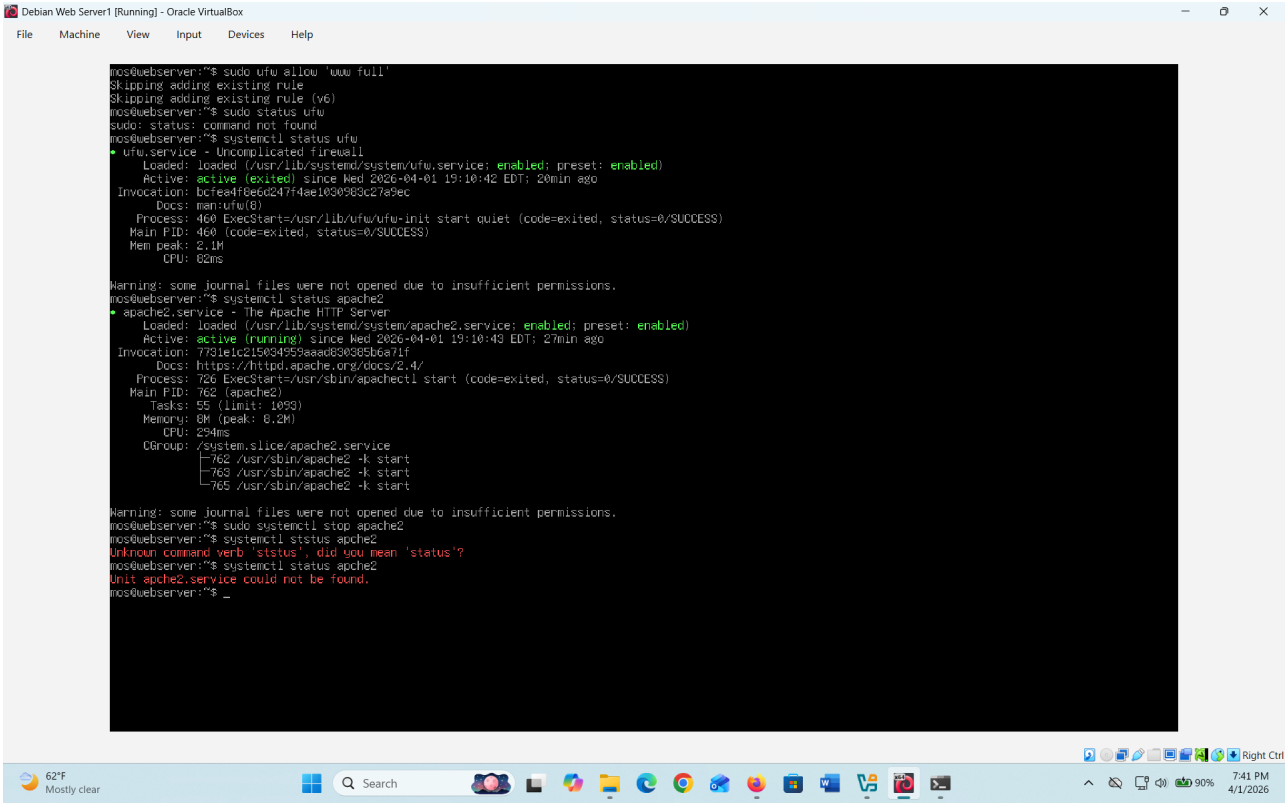
- **What is the command you use to stop Apache?**
- By using the command `sudo systemctl stop apache2`

- 

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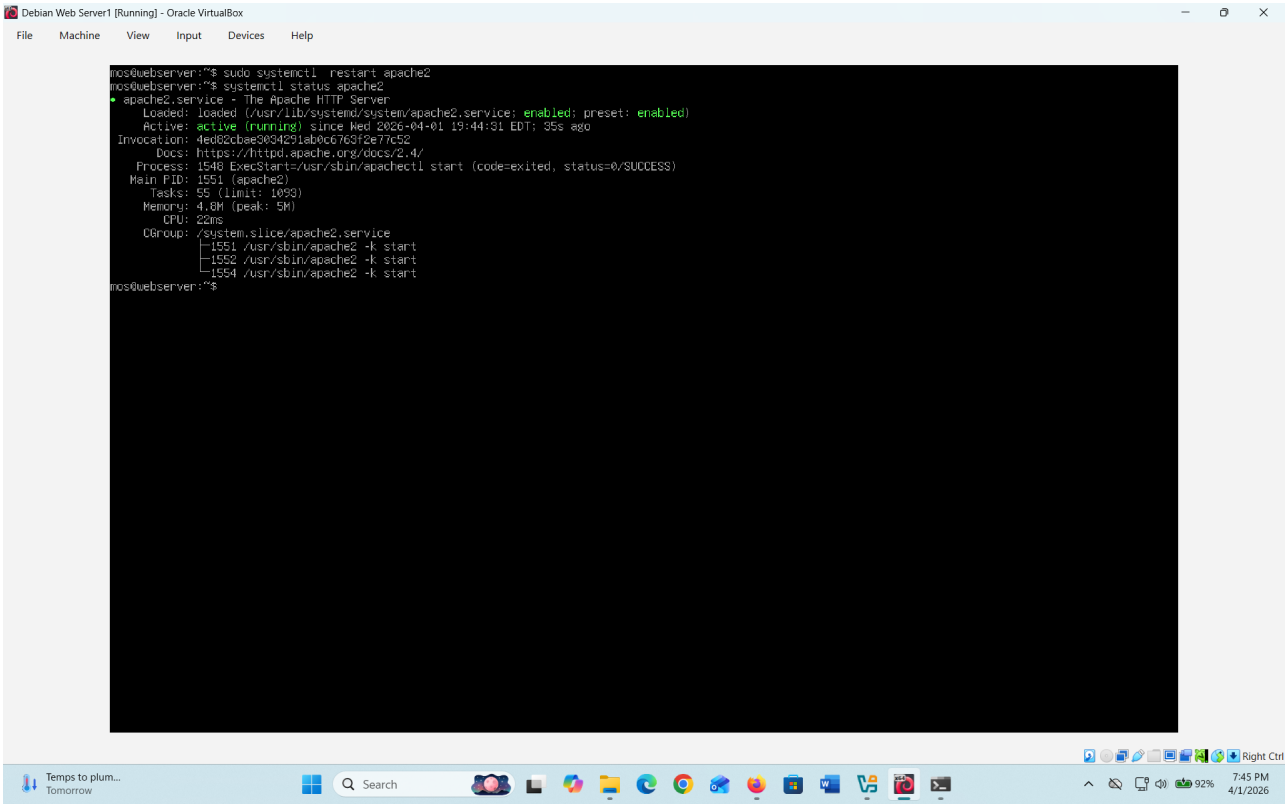
- **What is the command you use to restart Apache?**
- By using the command `sudo sytemctl restart aoache2`
- 

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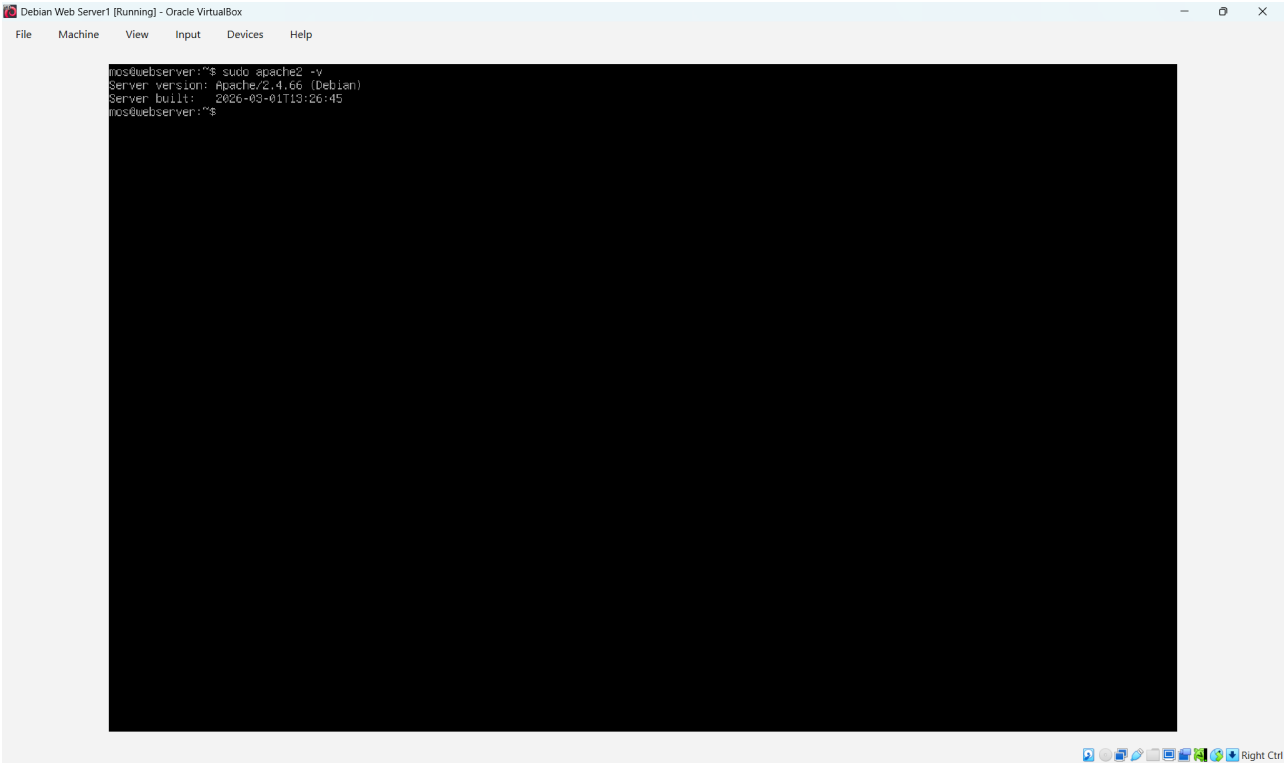
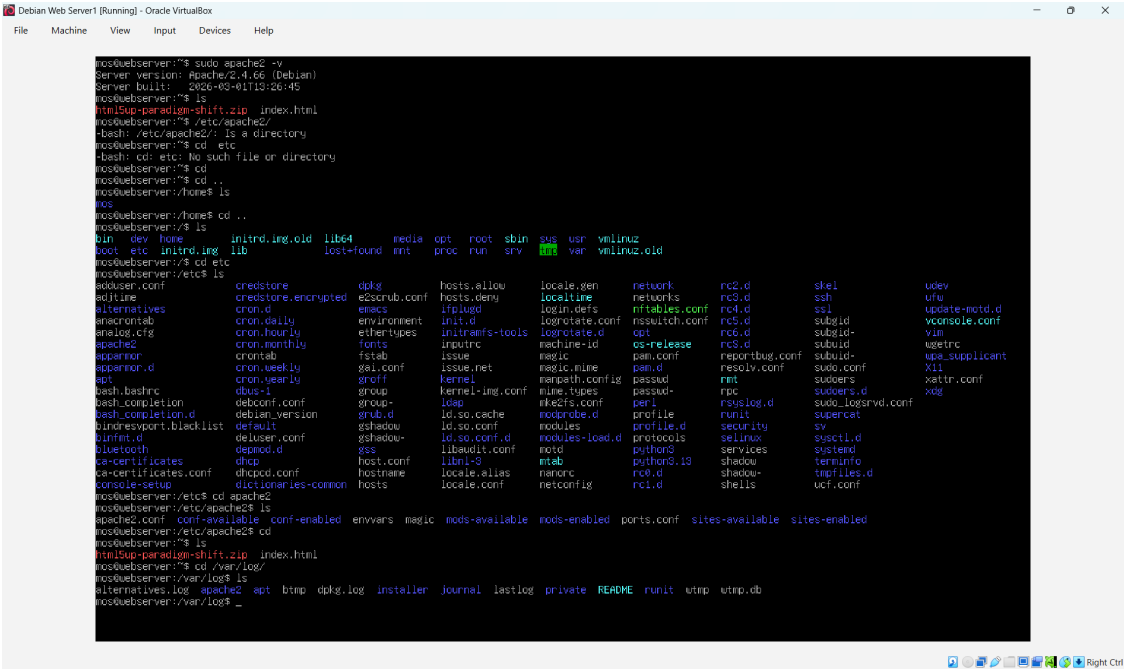
- **What is the command used to test Apache configuration?**
- By using the command `sudo apachectl configtest`

- **What is the command used to check the installed version of Apache?**
- By using the command `sudo apache2 -v`
- 

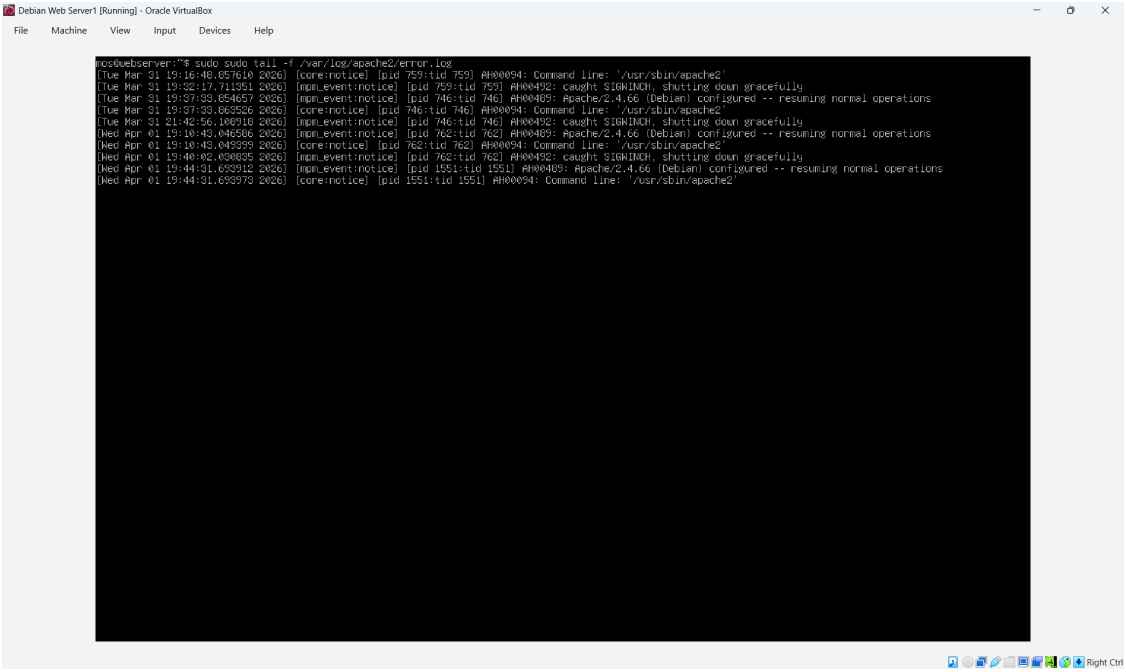
- **What are some common configuration files for Apache?**
    - **Main Configuration File**
    - `/etc/apache2/apache2.conf`
    - The core config file.
    - Controls global settings like:
    - Server defaults
    - Security rules
    - Directory permissions
- Other config files are included from here.
    - **Ports Configuration**
        - `/etc/apache2/ports.conf`
        - Defines which ports Apache listens on.
        - Example:
        - - Listen 80
        - - Listen 443
    - **Virtual Hosts**
        - `/etc/apache2/sites-available/`
        - Contains config files for each website (virtual host).
        - Example: 000-default.conf

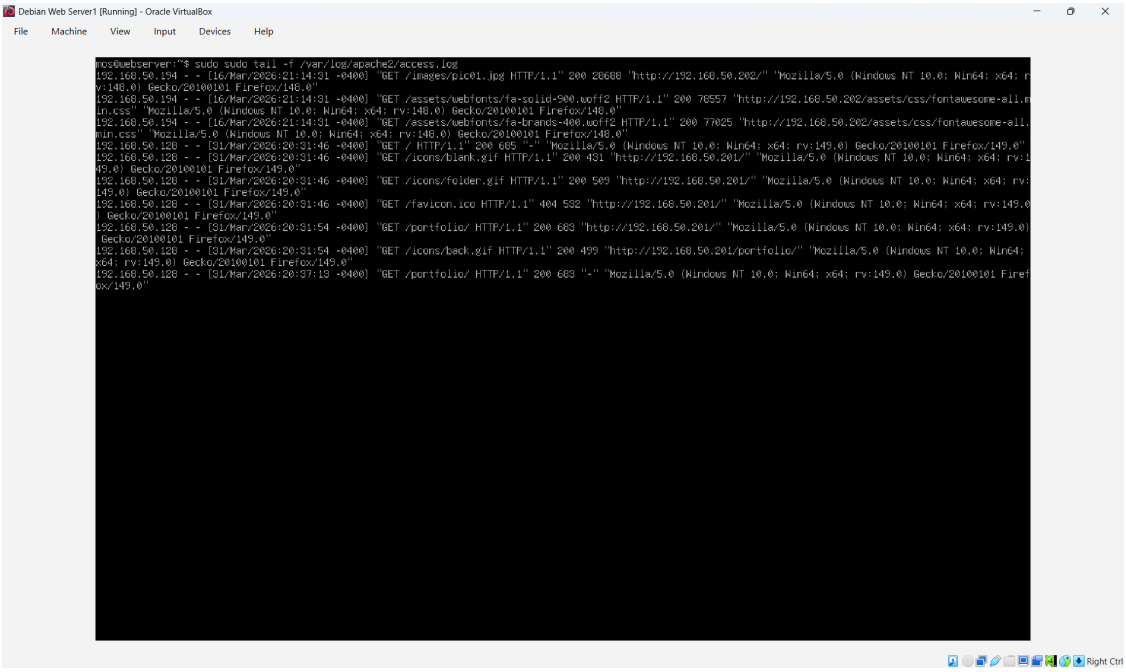
        - `/etc/apache2/sites-enabled/`
        - Symlinks to active sites.
        - Use:
        - a2ensite (enable site)
        - a2dissite (disable site)
    - **Modules Configuration**
        - `/etc/apache2/mods-available/`
        - All available modules (e.g., rewrite, ssl)
        - `/etc/apache2/mods-enabled/`
        - Active modules (symlinks)
        - Commands:
        - a2enmod rewrite
        - a2dismod rewrite
    - **Global Config Snippets**
        - `/etc/apache2/conf-available/`
        - `/etc/apache2/conf-enabled/`
        - Used for reusable config fragments (e.g., security settings, charset).
    - **Environment Variables**
        - `/etc/apache2/envvars`
        - Sets environment variables like:
        - Apache user/group (www-data) Log locations
- **Where does Apache store logs?**
    - **Default Log Directory**
    - `/var/log/apache2/`
    - This is where all Apache log files are kept by default.
    -
    - **Common Log Files**
        - Access Log
        - `/var/log/apache2/access.log`



- Records every HTTP request to your server.
- Includes:
  - Visitor Ip
  - Requested page
  - Response status (200, 404, etc.)

- **what are some basic commands we can use to review logs?**
  - **Viewing Logs**
    - ***cat – Display the whole file***
    - `cat /var/log/apache2/access.log`
    - `cat /var/log/apache2/error.log`
    - Good for small logs, not recommended for large files.
    - ***less or more – Scroll through files***
    - `less /var/log/apache2/error.log`
    - `more /var/log/apache2/access.log`

- 

- 

- Allows scrolling up/down.
- less is better: supports search with /term.